



**Cite this article:** Völler J, Linde-Domingo J, Ortiz-Tudela J, González-García C. 2026 From sudden perceptual learning to enduring engrams: a representational perspective. *Phil. Trans. R. Soc. B* **381**: 20250245.  
<https://doi.org/10.1098/rstb.2025.0245>

Received: 11 September 2025

Accepted: 5 February 2026

One contribution of 18 to a theme issue ‘The role of hippocampal predictions in cognition: bridging perception and memory’.

#### Subject Areas:

cognition, neuroscience, behaviour

#### Keywords:

one-shot perceptual learning, representations, long-term memory, prediction error, multiple traces, perceptual insight

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# From sudden perceptual learning to enduring engrams: a representational perspective

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Sudden perceptual learning refers to the abrupt disambiguation of an initially ambiguous stimulus. Despite the limited encoding time, these brief perceptual experiences transform into enduring memory engrams. Still, a comprehensive framework of why these memories are so long-lasting and their neural underpinnings, in particular the role of the hippocampus, is currently lacking. Here, we build on the apparent connection of sudden perceptual learning with long-term memory to outline how a representational perspective may help to overcome current limitations, similar to other areas of memory research. In short, we claim that sudden perceptual learning triggers initial prediction errors across the hierarchy, including strong higher-level (semantic) ones. Once a solution is found, the formerly ambiguous image can be connected to previous schemas and a disambiguation cascade of prediction error minimization is triggered, starting with semantic prediction errors. As a consequence, these events lead to low-dimensional representations that show signs of early semanticization, thereby favouring semantics over perceptual details. In sum, we propose that multiple memory traces are simultaneously encoded during sudden perceptual learning, and while low-dimensional representations engage neocortical structures, high-dimensional, episodic representations require the hippocampus. This perspective offers a promising framework to advance research on sudden perceptual learning and related phenomena.

This article is part of the theme issue ‘The role of hippocampal predictions in cognition: bridging perception and memory’.

## 1. Introduction

Perceptual learning, broadly defined as discovering a new structure or relation in sensory input, typically unfolds gradually through repeated exposure and extensive training [1–3]. In contrast to that, a special form of perceptual learning that operates on initially ambiguous input can occur rapidly (e.g. suddenly grasping the words of a song after reading the lyrics). Inside the laboratory, these types of phenomena can be recreated with Mooney stimuli ([4]; figure 1, left panel, *Input* row ‘pre’). Following just a single exposure to the clear version of that image (figure 1, left panel, *Input* row ‘disambiguation’), the formerly ambiguous stimulus can suddenly be disambiguated, reflected in higher recognition rates, object naming accuracy and faster reaction times [7–10]. Remarkably, despite the absence of repeated exposure, this sudden perceptual learning produces unusually long-lasting memories: disambiguated Mooney images can be vividly recognized for up to

three weeks [11] and accurately named for up to five months after encoding [12].

How are these long-lasting memories supported neurally? The hippocampus, given its well-established role in fast encoding of episodic information [13,14], has been proposed as a key structure in these processes (e.g. [15]). While there is robust evidence of hippocampal activation *during* sudden (perceptual) learning [11,16–20], evidence for its involvement in the *subsequent remembering* of the newly acquired information remains inconclusive. For instance, two previous studies established a direct role of the hippocampus in subsequent memory [21,22], with bilateral anterior hippocampus activity during Mooney image disambiguation predicting later memory for these images [21]. Conversely, several other studies report no significant correlation between hippocampal activity and subsequent memory and instead highlight the role of medial prefrontal cortex, amygdala and temporo-parietal structures [11,16,17]. Hence, a comprehensive understanding of how this phenomenon is implemented neurally, particularly with respect to the role of the hippocampus, is currently lacking. More broadly, how sudden perceptual disambiguation leads to such enduring memory traces remains unclear.

Combining the ‘insight-as-prediction error’ hypothesis [23], which links the enhanced memory to a sharp reduction in prediction error during abrupt shifts in understanding (as observed in sudden perceptual learning), with advances in other areas of long-term memory research, we aim to contribute a representational perspective to explain the unique mnemonic consequences of sudden perceptual learning. In particular, we suggest that this perspective, which has crucially contributed to better understanding of how and which information is encoded and retrieved over time (e.g. long-term memory: [24]; working memory: [25,26]), could offer new insights into the brain regions involved and the durability of the resulting memory traces. We begin by outlining the computations that might support these enduring memories. We then identify two potential limitations in existing research on sudden perceptual learning, before illustrating how a representational perspective could help to reconcile diverging findings and offer a more unified account of the phenomenon.

## 2. The role of prediction errors in sudden perceptual learning

The predictive processing framework has been influential in understanding the computations involved in sudden perceptual learning. This framework broadly states that perception is an active process of iterative inference, in which the brain generates predictions about the most probable causes underlying the sensory input [27,28]. These predictions from higher-level areas are compared with sensory, feed-forward information and any mismatch (the prediction error) is propagated back up the hierarchy for further processing [29]. The underlying assumption is that the system’s main goal is to minimize the said prediction error [27,30].

In sudden perceptual learning, viewed through the lens of predictive processing, the ambiguous perceptual input initially generates a prediction error (or several, as we will argue later) that cannot be minimized at that stage [31]. Subsequently, this prediction error triggers a search for a simpler generative model that explains the data (*Bayesian model reduction*; [32,33]). The recent ‘insight-as-prediction error’ hypothesis suggests that, in contrast to other phenomena where this process is gradual, sudden perceptual learning is characterized by a rapid, larger-than-expected minimization in prediction error. This sharp minimization is underpinned by activation in hippocampal–prefrontal pathways as well as enhanced dopamine and noradrenaline levels that in turn facilitate memory encoding [23]. How exactly the prediction error is reduced is a subject of ongoing research (e.g. [34,35]; see Box 1).

### Box 1: Prediction error minimization in sudden perceptual learning.

To explore the role of higher- versus lower-level information in sudden perceptual learning, recent studies manipulated specific properties of the clear image. By doing so, they showed that the disambiguation effect is invariant to size but affected by orientation and position manipulations, suggesting that changes in (mid- and higher-level) perceptual properties affect successful prediction error minimization. Moreover, when replacing the clear image with another exemplar from the same object category, there was no disambiguation measured by correct naming of the image [34,35]. Note that these manipulations did not directly affect semantic feature accessibility during the presentation of the clear image, which limits interpretations about the necessity of semantic knowledge in disambiguation. Another study showed that subjective recognition post-disambiguation depends on both higher-level semantic and lower-level features—albeit these latter ones to a lesser extent [10].

We propose that prediction error minimization relies on successfully relating the conceptual and perceptual representations of the event, possibly suggesting that different prediction errors along the cortical hierarchy need to be connected with each other. This aligns with findings that perceptual priors lead to widespread changes in the cortical hierarchy [7–9,36]. More research is needed to fully elucidate this open question, preferably using multiple tasks and time-scales to test the effect of different representational contents on subsequent disambiguation.

Importantly, recent work calls for a better characterization of the different parameters of predictions and prediction errors, e.g. their mnemonic origin [37]. Similarly, influential models suggest that predictions are hierarchically organized within different memory systems and that the level at which the prediction error occurs influences which information is encoded (e.g. if an unexpected item occurs in a familiar context, this leads to a large residual hippocampal prediction error that drives episodic encoding of that item; [38]). In the context of one-shot declarative memory (i.e. the encoding of a singular scene–item pairing), there is evidence that cases that produce larger (hippocampal) prediction errors are subsequently recognized better, supporting the framework’s hypothesis [39].

Thus, we argue that without characterizing the content of said prediction errors and the representational consequences of minimizing them, current frameworks are limited in their explanatory power. Below we illustrate two of these limitations in more detail.

### 3. Limitations in current research

#### (a) Location over representational content

While there is a multitude of studies concerning the neural underpinnings of sudden perceptual learning (the ‘*where*’; e.g. [11,12,21]), substantially less research regards the representational consequences of these events (i.e. ‘*what*’ exactly is learnt). A methodological consideration illustrates the relevance of this distinction.

To probe memory from sudden (perceptual) learning events, studies have primarily used two tasks: a recognition task (e.g. ‘Is this stimulus old or new?’) and a solution retrieval task, whereby the participants are asked to re-solve the ambiguous stimulus (e.g. name the Mooney image). Interestingly, when memory is assessed based on *solution knowledge*, retrieval appears to rely primarily on prefrontal regions [11,16,17]. By contrast, when mainly *recognition* memory of the Mooney image is tested, the hippocampus shows stronger involvement ([12,21]; note that Becker *et al.* [21] only asked for solution retrieval for images that were subjectively recognized as old, making memory performance dependent on recognition).

Thus, there is no single answer to the ‘*where*’ of sudden perceptual learning. Instead, the involvement of a specific brain region will strongly depend on the representational content of the to-be-retrieved information. Specifically, recognition tasks access relatively more episodic information (i.e. spatio-temporal features of a memory), while solution knowledge is a more indirect measure tapping into semantic memory features (cf. [17,40,41]), supported by the hippocampus and neocortical areas, respectively.

This suggests that to understand the neural underpinnings of sudden perceptual learning, we need to characterize the representational content of what is learnt—a perspective that has already been highlighted in other fields of memory research [42,43]. We argue that by shifting the focus towards understanding the content of predictions and prediction errors during sudden perceptual learning, we can gain a deeper understanding of how these memories are formed and what exactly is being encoded during these episodes.

#### (b) One-engram perspective

Another important aspect that is often implicit in the field is the assumption of a ‘one-engram perspective’ (i.e. that a singular memory trace is encoded during sudden perceptual learning). Coined by Richard Semon, an *engram* refers to the enduring physical or chemical changes as a result of learning [44] and will be used here to denote any offline representation of a past experience [45]. This ‘one-engram perspective’ is reflected in methodological decisions made by researchers, which make it difficult to dissociate between potentially distinct memory traces, such as defining successful retrieval as the conjunction of recognition and solution memory (e.g. [21]) or using fixed task orders so that task one (e.g. resolution) has an influence on task two (e.g. old/new judgements), preventing independent measurement of these task performances (e.g. [21,46,47]).

Although the involvement of multiple memory systems has been acknowledged in sudden perceptual learning research [21,23,41], their interaction has likewise been characterized from a one-engram perspective. For instance, it has been proposed that one memory trace is encoded via the initial detection of a prediction error by the hippocampus and a subsequent schema-congruent integration via neocortical pathways [23], consistent with the mechanisms suggested in previous models (SLIMM; [48,49]).

However, a seminal lesion study critically questions the one-engram perspective. This study found that hippocampal-lesioned patients display comparably good solution memory performance up to five months after encoding, measured via naming the Mooney images, but they do not recognize (old/new question) the images one week after encoding. Moreover, they found that solution performance in healthy controls was facilitated at early intervals, possibly via available declarative knowledge from the encoding phase, but then declined over time, whereas it remained relatively stable in patients [12]. While these findings suggest that specific contents from sudden perceptual learning can be retrieved without the hippocampus (see also [11,16,17]), they more importantly indicate that different memory systems support the encoding and retrieval of different memory features during these episodes, conflicting with a one-engram perspective. Additionally, they suggest that these different memory features evolve differently over time. Given the scarce evidence in this regard, we believe that more research to elucidate the contribution of different memory systems to these effects is needed. Next, we show examples of long-term memory research that showcase the benefits of shifting the focus from location to representational contents.

### 4. The advantages of a representational perspective in memory research

#### (a) Representational content explains location

Similar to sudden perceptual learning, a large body of memory research has been centred on the identification of the neural substrate underlying this process in the brain [50–52]. This approach has made important contributions to the field, for instance,

mapping long-term declarative memory to the hippocampus and postulating that memory and perception, underpinned by the visual ventral stream, are distinct modules [53,54]. However, this process-to-location mapping idea could not account for the variety of empirical evidence, showing hippocampal involvement in high-level perceptual processing [55–57] and visual ventral stream in mnemonic processing [58,59].

Thus, more recent proposals postulate that the distinctions between brain regions might be better explained by the representational content on which they operate or by the computations that they applied to this content [42,43,60]. Along those lines, there is evidence that while high-dimensional representations (e.g. the conjunction of complex elements) require the hippocampus, low-dimensional representations (e.g. simple, single-item information) can be encoded by neocortical structures (see [42] for an overview), above and beyond traditional process labels. As such, the representational perspective has enabled a better understanding of mnemonic dynamics, including the involvement of the underlying structure. Notably, this approach is not exclusive to long-term memory research but has also been fundamental in studying other memory systems, such as working memory (e.g. [61]).

## (b) Multiple-engram perspective

Another aspect that was crucially advanced through a representational perspective regards how memories are consolidated. Comparable to current proposals in sudden perceptual learning, one of the first computational models of system consolidation—the complementary learning system theory [62,63]—built on a one-engram perspective. Specifically, the framework proposed that a single memory engram is rapidly encoded in the hippocampus and is transferred to the neocortex after a specific amount of time. However, several more recent studies provide evidence for (i) rapid neocortical engrams without hippocampal involvement [64–66] and (ii) more remote memories being affected after hippocampal lesions [67–69]. These studies disrupt the rigid division between two learning systems that operate on the same engram but at different points in time.

Instead, new theories propose a multiple-engram perspective by which multiple psychological and neurobiological representations are simultaneously encoded by different brain regions during an event, ranging from detail-rich (by hippocampal structures) to more gist-like and semantic representations (in neocortical structures) [70–72]. In this regard, there is compelling evidence for a loss of details in favour of gist-abstraction as memories age, termed *semanticization* [24,73–76]. Although the semanticization of memory as a form of dimensionality reduction is important for everyday cognition (for a detailed discussion on the mechanisms of dimensionality transformations see [77]), recent studies have shown that encoding and retrieval demands play an important role in how memories are stored [78,79], and that the strengthening of conceptual information does not necessarily come at the expense of episodic details [80].

Altogether, these examples show how a representational perspective has advanced our understanding of long-term memory. Next, we explain how these approaches can be applied to sudden perceptual learning.

## 5. From where to what: the role of representations in sudden perceptual learning

When faced with an ambiguous stimulus, participants' main task is to disentangle or disambiguate that stimulus (figure 1, left panel, *Input* row). As proposed elsewhere, once disambiguated, i.e. when the initial prediction error is suddenly reduced, specific prediction-error-driven mechanisms at the systems and molecular levels enhance memory encoding ('insight-as-prediction error'; [23]). We argue that this initial ambiguity induces prediction errors on different levels of the representational hierarchy, including lower-level sensory and higher-level semantic prediction errors, and that characterizing the representational content of these errors is essential for understanding their mnemonic consequences. For instance, higher-level prediction errors arise because the observer initially thought that the Mooney image (figure 1, *Input* row) depicts a 'horse' instead of a 'bee', while perceptual prediction errors entail the image's lower-level perceptual properties that cannot be recognized (e.g. edges, orientations, shadows and colours). Note that these ideas generally align with other models of perceptual ambiguity (e.g. 'analysis-by-synthesis', [81]), which would assume that the currently available low-level cues cannot activate a high-level generative model, and that in turn the prior knowledge fails to approximate the perceptual input [31].

We propose that a sudden resolution of the higher-level semantic prediction error leads to a disambiguation cascade (figure 1, left panel, *Prediction Error* row), whereby the prediction errors in all other levels are minimized consequently (see §7 Open questions), broadly following Bayesian inference accounts. While we assume continuous bidirectional constraints between levels (i.e. low-level cues informing semantic hypotheses and *vice versa*; see Box 1), the term 'cascade' refers here to the net temporal order of resolution: the moment of high-level resolution precipitates a rapid, downward stabilization and integration of prediction errors across the hierarchy. This integration would manifest phenomenologically in an 'Aha!' experience (Box 2). Crucially, the newly acquired (meaningful) information can be easily connected to previous schemas as it fits with prior knowledge [23,41].

### Box 2: The Aha! experience in sudden perceptual learning and its influence on memory.

The experience of an insight that subjectively manifests in feelings of happiness, surprise and certainty in the solution ('Aha!') is a frequent phenomenon in sudden perceptual learning. Recent frameworks relate the Aha! experience to the aforementioned sudden reduction in prediction error that is followed by an increase in dopaminergic precision-weighting



[31] and positive metacognitive prediction errors [23]. These ideas are corroborated by behavioural [47,82–84] and neural evidence, for instance, linking the Aha! to increased hippocampal activity [19,21].

Here, we extend this idea to highlight the importance of reducing and integrating all initial prediction errors on different representational levels to produce an ‘Aha!’. Studies similar to those mentioned in Box 1 could explore the influence of different representational features on the Aha! experience. Subsequently, as this integration is inherently episodic (and rewarding), we argue that the Aha! experience acts as a ‘hippocampal booster’, triggering the hippocampus to encode a new engram that contains detailed event information (cf. [85]). We therefore expect correctly solved trials with stronger versus weaker Aha! levels to benefit episodic memory but not necessarily semantic memory. Initial evidence for this hypothesis comes from studies showing that the Aha! experience enhances memory for irrelevant stochastic facts presented in close temporal succession [86].

Note that the high- versus low-insight comparison is fundamentally different from simply contrasting successful versus unsuccessful disambiguation, which assumes an equivalent effect of experiencing an insight and successful disambiguation. Studies using the former comparison (e.g. [21]) show hippocampus-based activity, in line with the notion of Aha! as a hippocampal booster, while the latter do not [11,16,17].

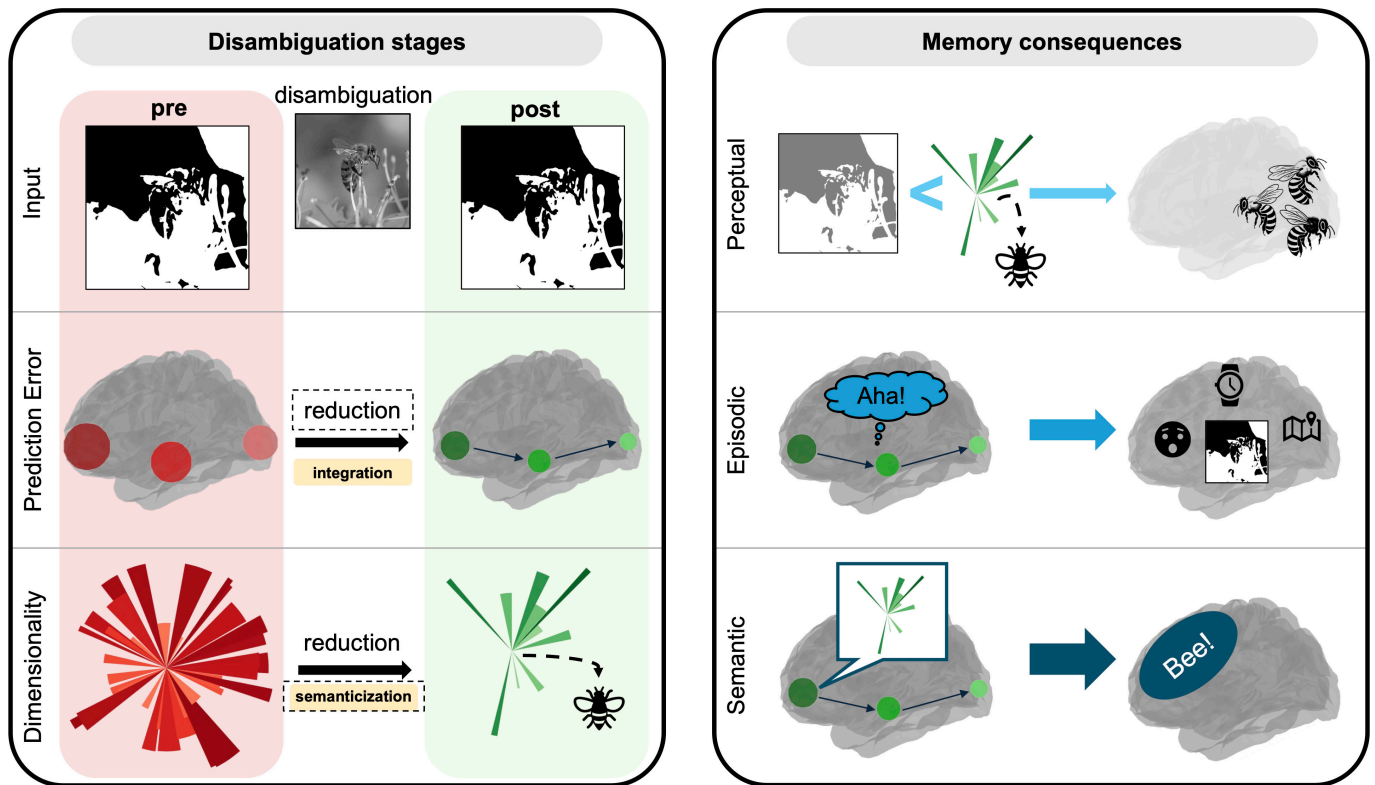
Two main representational consequences follow. On the one hand, the sudden reduction of semantic prediction errors results in a strong encoding of semantic information from the event (cf. [38]; figure 1, right panel, *Semantic* row). Simultaneously, this unexpected reduction and critically integration of (not only higher- but also mid- and lower-level) prediction errors is inherently episodic. As such, it is experienced as a rewarding and salient event (the ‘Aha!’; see Box 2), thereby specifically boosting the encoding of episodic information (i.e. detail-rich temporo-spatial memory) over other representational formats (figure 1, right panel, *Episodic* row). On the other hand, the now-available connection with pre-existing schemas reduces the dimensionality of the to-be-encoded information (i.e. explaining away the input; [87,88]; figure 1, left panel, *Dimensionality* row). This, in turn, results in a low-dimensional representation that weights conceptual information more strongly than perceptual details, resembling an early instance of memory semanticization, evident in comparably weaker memory for perceptual details of the episode (figure 1, right panel, *Perceptual* row). We argue that the rapid minimization of prediction errors, which leads to an episodic boost, paired with an early semanticization in higher-level representations afforded by the established perceptual priors explain the longevity of the memory traces.

In summary, we propose that multiple memory traces are encoded simultaneously during sudden perceptual learning, with a strong reliance on semantic and episodic information but, and perhaps counterintuitively, a lesser focus on perceptual information. Comparing unsuccessful with successful disambiguation trials can uncover the specific consequences of sudden perceptual learning. We argue that only successful disambiguation trials entail this rapid reduction in prediction errors and semanticization of the input. As a consequence, we predict better semantic and episodic memory for successful compared with unsuccessful disambiguation trials but comparable perceptual memory performance across both conditions (note that, critically, non-disambiguated Mooney images cannot be semanticized). Experimentally, disambiguation can be manipulated via either presenting the correct clear image in between the Mooney images (see figure 1, *Input* row), or presenting a different, non-corresponding clear image, reducing the probability of disambiguation. Neurally, we suggest a complementary interplay between hippocampus and neocortex to underlie these effects, optimally balancing generalizability (low dimensionality) and separability (high dimensionality; [89]). Specifically, the hippocampus encodes high-dimensional episodic details from the event (e.g. [90]), boosted by the ‘Aha!’ experience. At the same time, prefrontal regions encode more abstract, semantic contents that induce a compression of the dimensionality of neural representations in these regions (e.g. [91,92]). Moreover, we predict that the dimensionality in the neural activity decreases from pre- to post-disambiguation in prefrontal regions, while it simultaneously increases in the hippocampus. In turn, we hypothesize that this balance will predict behavioural performance in tasks relying on semantic (e.g. naming) and episodic contents (e.g. recognition, spatio-temporal aspects), respectively.

Previous studies provide initial evidence for some of these ideas in isolation. After successful disambiguation, Mooney images display a representational shift towards their unambiguous version throughout the cortical hierarchy (i.e. the activity patterns induced by post-disambiguation Mooney images are more similar to those induced by the clear image and less similar to the same Mooney image pre-disambiguation; [8]). This finding suggests that the memory traces show stronger weighting of prior information over the specific sensory input, potentially making them more prone to early semanticization. Moreover, studies have shown that prior-knowledge-aided processing generally benefits memory encoding but potentially at the cost of discriminating perceptual details of the memory [93,94]. This supports the idea that successful disambiguation entails benefits for higher-level but not necessarily lower-level perceptual details.

## 6. Future directions

Testing our hypothesis necessitates a multi-task multi-timescale approach. We therefore call for a testing battery that taps into different representational features of the memory, which we predict to involve different brain regions. When probing detail-rich contextual information from a sudden perceptual learning episode, the hippocampus will demonstrate subsequent memory effects. By contrast, when probing higher-level semantic information that is less complex in its dimensionality, non-hippocampal areas such as the prefrontal cortex will be recruited. Over time we predict an increased involvement of cortical regions to the detriment of the hippocampus as perceptual and episodic details are forgotten.



**Figure 1.** An integrative overview of sudden perceptual learning from a representational perspective, encompassing elements from previous theories (dashed lines) and predictions derived from the framework proposed here. **Left panel: Input row.** Actual input to the visual system. The Mooney image is presented twice, once before (pre) and once after (post) disambiguation, induced by its unambiguous clear version (disambiguation). **Prediction Error row.** The Mooney image at pre-disambiguation induces prediction errors across the hierarchy with a strong higher-level, semantic error. These prediction errors are suddenly reduced at post-disambiguation following a temporal gradient from higher-level to lower-level errors with the help of newly acquired knowledge from the clear image, resulting in an integration of the prediction errors. **Dimensionality row.** The Mooney image's representation is drastically reduced in its dimensionality from pre- to post-disambiguation as the image is connected with previous schemas, and its representation favours conceptual over perceptual information (*semanticization*). **Right panel: Perceptual row.** As a result of the dimensionality reduction, the actual perceptual input is weighed less (<) than the disambiguated and semanticized version of the Mooney image, resulting in a decreased encoding of its perceptual details (e.g. the original percept's specific orientation or colour) in posterior brain regions. **Episodic row.** The sudden reduction and integration of prediction errors across the hierarchy constitutes a salient and rewarding event, subjectively experienced as an 'Aha!' that boosts the encoding of episodic formats (e.g. spatio-temporal or contextual information) in the medial temporal lobe. **Semantic row.** Finally, both dimensionality reduction and semanticization facilitate a strong mnemonic encoding of semantic information (e.g. object identity, category information) in neocortical areas. Mooney and clear images in the figure are extracted from THINGS-Mooney [5], which is based on THINGSplus [6] and are copyright-free.

How memory traces semanticize could furthermore elucidate the neural architecture underlying these memories. Different features can either be stored within the same 'deep' multilayered trace [24,95] or as multiple traces with partially independent decay functions [96,97]. As the distinction between these memory architectures remains an active focus of research, our predictions are necessarily of speculative nature. In multilayered accounts, the selective strengthening of one task-relevant feature, e.g. conceptual information, is likely to reduce the relative accessibility of task-irrelevant features, e.g. perceptual details ([74]; note that perceptual features can also be selectively enhanced; [95]). By contrast, semanticization in multiple memory traces may be less constrained by task demands and by other features' states. If sudden perceptual learning conforms to a multilayered architecture, we therefore predict that stronger semanticization owing to larger semantic prediction errors would result in a stronger reduction in perceptual details. Moreover, we predict that long-term stable disambiguation involves a reduction and integration of all (and not only some) prediction errors, which could be tested by manipulating which levels of prediction error can be resolved during disambiguation (see Box 1). If this reduction follows predictive processing accounts, we predict that higher-level areas encode recognition-related information earlier than lower-level regions [98] and that this information is communicated from higher-level deep layers to agranular lower-level layers [99].

Regarding dimensionality, a wide range of studies has used Principal component analysis-based or decoding-based approaches (e.g. 'shattering dimensionality') that seem to track relatively well with intuitions about the dimensionality of representational formats, e.g. higher dimensionality for more complex representations [89,100,101]. Importantly, previous studies suggest that dimensionality is behaviourally relevant. For instance, higher-dimensional representations in the hippocampus predict better episodic memory [90] and lower-dimensional representations in medial prefrontal regions facilitate abstraction and knowledge generalization (see [102] for an overview). These advances can be used to explore the influence of dimensionality transformations on specific behavioural tasks. A concurrent characterization of the content that spans the dimensionality using decoding and representational similarity approaches is likewise important [89].

The present framework has been discussed here in the context of sudden visual learning. However, we believe that the proposed disambiguation cascade can be triggered by any stimulus or any relationship between stimuli that is initially ambiguous on several levels of the representational hierarchy (e.g. perceptually and semantically ambiguous) and whose

resolution carries a drastic semantic change. Moreover, this ambiguity needs to be resolved suddenly, e.g. via implicit restructuring of the problem space, internal Bayesian model reduction or providing a specific cue. Previous studies have shown that the facilitating memory effects of sudden learning disappear when the learnt association is (semantically) arbitrary, highlighting the importance of the semantic connection [103]. We therefore believe that this framework could for instance account for the reported effects in speech-in-noise scenarios (e.g. [104,105]), sudden vocabulary learning (e.g. [106]) and resolving ambiguity in abstract art (e.g. [107,108]).

Finally, this framework yields testable developmental predictions for the proposed mechanisms. Previous literature has shown that children's sudden visual learning ability is limited [109,110], possibly owing to immature prior knowledge structures [111] and a more feature-based processing strategy [112]. As a result, children may generate weaker semantic prediction errors and thus show less dramatic error reduction, leading to a weaker Aha! experience and memory traces that remain more perceptual than semantic. By contrast, older adults may rely more strongly on semantic memory, which could shape their learning and memory integration in more semanticized ways. A previous study manipulating the level of prediction errors in action schemas showed that prediction error-driven encoding is more important in young children, while schema-driven encoding shapes memory in older adults ([113]; see also [114]).

In this opinion piece, we have explored how a representational perspective can help us better understand the multifaceted phenomenon of sudden perceptual learning. We especially highlighted the need to characterize the content of the prediction errors induced by the ambiguous perceptual input. These prediction errors are suddenly reduced in a disambiguation cascade, resulting in a lower-dimensional representation that is easy to integrate with previous schemas, thus promoting stable, long-term encoding and a simultaneous boost of episodic information as this sudden prediction error minimization constitutes a salient and rewarding event. Altogether, we hope that this perspective inspires new research into the representational architecture of sudden perceptual learning, helping to characterize the features of these unique memories.

## 7. Open questions

- (i) Does the reduction of prediction errors follow a temporal sequence, as suggested here? For example, are semantic prediction errors resolved first, followed by mid-level and then lower-level errors? Moreover, is the reduction of these prediction errors dependent on each other conforming to a sort of temporal hierarchy?
- (ii) Regarding subsequent memory, are all prediction errors equally important in remembering information from the perceptual stimulation? Does the importance of reducing one prediction error depend on the type of information to be retrieved?
- (iii) How do the memory traces derived from disambiguation interact? Does the influence of one memory trace on the other remain stable over time?
- (iv) What are the main representational components of sudden perceptual learning, and does this framework extend beyond perceptual, episodic and semantic representations to include additional components?

**Ethics.** This work did not require ethical approval from a human subject or animal welfare committee.

**Data accessibility.** This article has no additional data.

**Declaration of AI use.** AI was used to ensure the readability and language of this work.

**Authors' contributions.** J.V.: conceptualization, project administration, visualization, writing—original draft, writing—review and editing; J.L.-D.: conceptualization, funding acquisition, supervision, writing—review and editing; J.O.-T.: conceptualization, funding acquisition, supervision, writing—review and editing; C.G.-G.: conceptualization, funding acquisition, project administration, supervision, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

**Conflict of interest declaration.** We declare we have no competing interests.

**Funding.** J.V. was supported by an FPI grant CEX2021-001161-M-20-5 funded by MICIU/AEI/10.13039/501100011033 and by FSE+. J.L.-D. was supported by Project PID2023-151104NA-I00 funded by MCIN/AEI/10.13039/501100011033 and by FEDER, EU, and Grant RYC2021-033940-I funded by MCIN/AEI/10.13039/501100011033 and by the European Union NextGeneration EU/PRTR. J.O.-T. was supported by an European Research Council Starting Grant (ERC-2024-StG-CONNECTS-101161992) and Grant RYC2023-045452-I funded by MICIU/AEI/10.13039/501100011033 and by the FSE+. C.G.-G. was supported by Project PID2023-149428NB-I00 funded by MCIN/AEI/10.13039/501100011033 and by FEDER, EU, and Grant RYC2021-033536-I funded by MCIN/AEI/10.13039/501100011033 and by the European Union NextGeneration EU/PRTR. The Mind, Brain and Behavior Research Center receives funding from grants CEX2023-001312-M by MICIU/AEI/10.13039/501100011033 and UCE-PP2023-11 by the University of Granada. The funders had no role in the preparation of the manuscript. This manuscript is part of the PhD thesis of J.V.

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